DISCOUNT FACTOR AS A REGULARIZER IN REINFORCEMENT LEARNING

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REINFORCEMENT LEARNING
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OVERVIEW

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- Theoretical background

2 Methodology

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- Uniformity ditribution
- Mixing time rate

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- Deep RL
- Windy gridworld

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PROBLEM

Reinforcement Learning algorithms often **struggle to generalize**, particularly when trained on limited data.

GOAL

Provide a formal theoretical characterization of indirect regularization induced by using a discount factor lower than the one specified by the task.

METHODOLOGY

Study TD-based algorithms with finite data, prove the equivalence between reduced discount and activation regularization, and compare discount regularization with explicit L_2 regularization.

THEORETICAL BACKGROUND

Markov Decision Process

$$\mathcal{M} = (S, A, P, R, \mu, \gamma)$$

- Evaluation discount γ_e: true objective
- Guidance discount γ: learning dynamics

Discounted Value Function

$$V_\gamma^\pi(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s\right]$$

We focus on a TD(0) learning in a batch setting to make the equivalence more transparent:

$$\theta_{i+1} := \theta_i + \alpha_i \left(\xi r + \gamma \hat{V}_{\theta_i}(s') - \hat{V}_{\theta_i}(s)\right) \nabla \hat{V}_{\theta_i}(s) - \alpha_i \nabla(\Psi)$$

- Relies on semi-gradient updates, the target $\hat{V}_{\theta_i}(s')$ is fixed
- A diminishing step size $\alpha_i = \frac{1}{i}$ is used
- Finite dataset: $D = (s, a, r, s')$

PROPOSITION 1

Let $\theta_1, \theta_2, \ldots$ parameters produced by the Batch TD(0) using a discount factor $\gamma < \gamma_e$, with $\xi = 1, \Psi \equiv 0$, initial parameters θ_0 and learning rate α_i. The algorithm produces the same sequence of parameters $\theta_1, \theta_2, \ldots$ if it is run with the discount factor $\gamma = \gamma_e$, but with added regularization function $\Psi(s, \theta) := \lambda(\hat{V}_\theta(s))^2$, $\lambda := \dfrac{\gamma_e - \gamma}{2\gamma}$, reward scaling $\xi := \dfrac{\gamma_e}{\gamma}$, learning rate $\alpha'_i := \dfrac{\gamma}{\gamma_e}\alpha_i$ and the same initial parameters θ_0.

Learning with $\gamma < \gamma_e$ and no explicit regularization is equivalent to learning with $\gamma = \gamma_e$ and an added regularization term on the estimated values.

EQUIVALENT HYPER-PARAMETERS

REGULARIZATION	REWARD SCALING	LEARNING RATE
$\lambda = \dfrac{\gamma_e - \gamma}{2\gamma}$	$\xi = \dfrac{\gamma_e}{\gamma}$	$\alpha'_i := \dfrac{\gamma}{\gamma_e}\alpha_i$

DEMONSTARTION *of Proposition 1*

$$
\begin{aligned}
\theta_{i+1} &= \theta_i + \alpha_i \left(r + \gamma \hat{V}_{\theta_i}(s') - \hat{V}_{\theta_i}(s) \right) \nabla \hat{V}_{\theta_i}(s) && (1.1)\\
&= \phi_i + \alpha_i \left(r + \gamma \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) && (1.2)\\
&= \phi_i + \alpha_i \frac{\gamma}{\gamma_e} \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \frac{\gamma_e}{\gamma} \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) && (1.3)\\
&= \phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) + \hat{V}_{\phi_i}(s) - \frac{\gamma_e}{\gamma} \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) && (1.4)\\
&= \phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) - \alpha_i' \frac{\gamma_e - \gamma}{\gamma} \hat{V}_{\phi_i}(s) \nabla \hat{V}_{\phi_i}(s) && (1.5)\\
&= \phi_i + \alpha_i' \left(\frac{\gamma_e}{\gamma} r + \gamma_e \hat{V}_{\phi_i}(s') - \hat{V}_{\phi_i}(s) \right) \nabla \hat{V}_{\phi_i}(s) - \alpha_i' \nabla \left(\frac{\gamma_e - \gamma}{2\gamma} \left(\hat{V}_{\phi_i}(s) \right)^2 \right) && (1.6)\\
&= \phi_{i+1} && (1.7)
\end{aligned}
$$

ACTIVATION REGULARIZATION

Thanks to Proposition 1, it is possible to analyze the regularization nature.
Such formulation allows to regularize large value predictions and improve generalization.
N.B. it does not penalize the parameters, but the estimated value function!

From Proposition 1, the penality term is defined as $\lambda = \dfrac{\gamma_e - \gamma}{2\gamma}$:if γ descreases, then λ increases.

To better understand the regularization, let's consider the linear case $\hat{V}_\theta(s) = \phi(s)^\top \theta$ and assume that the features are orthogonal under the sampling distribution, that is $\mathbb{E}_{s\sim D}\left[\phi(s)\phi(s)^\top\right] = \Lambda$
It is possible to re-write the activation regularization term as:

$$\lambda\, \mathbb{E}_{s\sim D}\left[\left(\hat{V}_\theta(s)\right)^2\right] = \lambda\, \mathbb{E}_{s\sim D}\left[(\theta^\top \phi(s))^2\right] = \lambda\, \theta^\top \Lambda \theta$$

The matrix Λ is the second-moments matrix of the features. The crucial point is that the regularization is not equal, instead it is governed by the distribution of the state space.

UNIFORMITY OF THE DISTRIBUTION

The term $\lambda \mathbb{E}_{(s)\sim D}\left(\hat{V}_\theta(s)\right)^2$ highlights the role of the empirical state distribution.

In the tabular case, activation regularization reduces to $\lambda \mathbb{E}_{(s)\sim D}\theta^2$, so rarely visited (or unseen) states receive little or no regularization.
Therefore, activation regularization is more effective when the empirical state-visitation distribution is close to uniform.

In experiments, this is addressed by sampling from a uniform policy in the tabular setting.
To measure how close the empirical distribution d_D is to the uniform distribution $u(s) = 1/|\mathcal{S}|$ the total variation distance is computed:

$$\mathrm{TV}(d_D, u) = \frac{1}{2}\sum_{s\in\mathcal{S}} |d_D(s) - u(s)| = \frac{1}{2}\sum_{s\in\mathcal{S}} \left| d_D(s) - \frac{1}{|\mathcal{S}|} \right|$$

MIXING TIME RATE

The mixing time describes **how quickly** the state distribution approaches the stationary distribution. It can be estimated by the inverse of the **spectral gap**:

$$\tau_{\text{mix}} \propto \frac{1}{\text{gap}(P)}, \qquad \text{gap}(P) = 1 - |\lambda_2(P)|$$

SLOW MIXING

A large τ_{mix} implies slow mixing: trajectories remain confined to restricted regions of the state space, producing correlated and less representative samples. In activation regularization, the penalty is

$$\lambda \mathbb{E}_{s \sim D}\left[\hat{V}_\theta(s)\right]$$

Frequently visited states are strongly regularized, while rare or unseen states receive little or no regularization. With slow mixing, coverage becomes uneven and estimation variance increases; in this regime, stronger regularization is typically beneficial.

TABULAR

Environment: 4×4 GridWorld
Policy Evaluation: Batch-TD, LSTD
Policy Optimization: SARSA
Evaluation: $\left\|\hat{V} - V^{\pi}_{\gamma_e}\right\|_2$,
where $\gamma_e = 0.99$

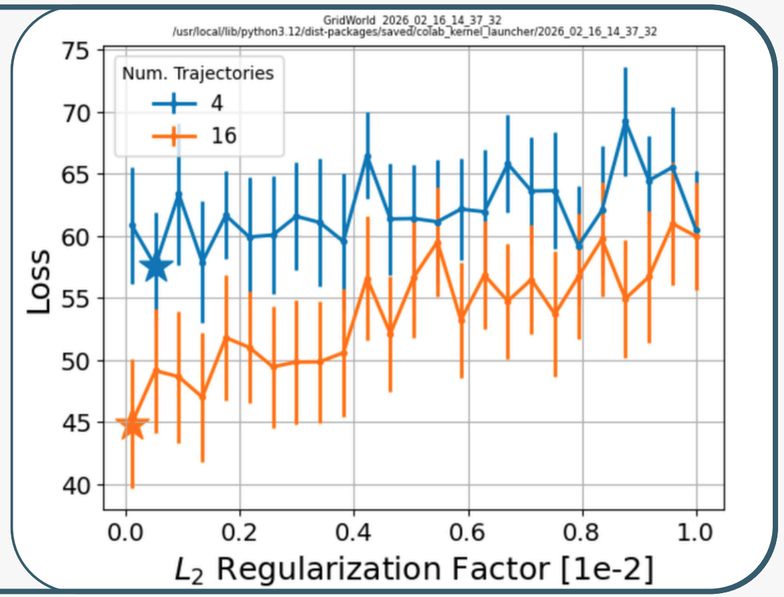
EFFECT OF DATASET SIZE

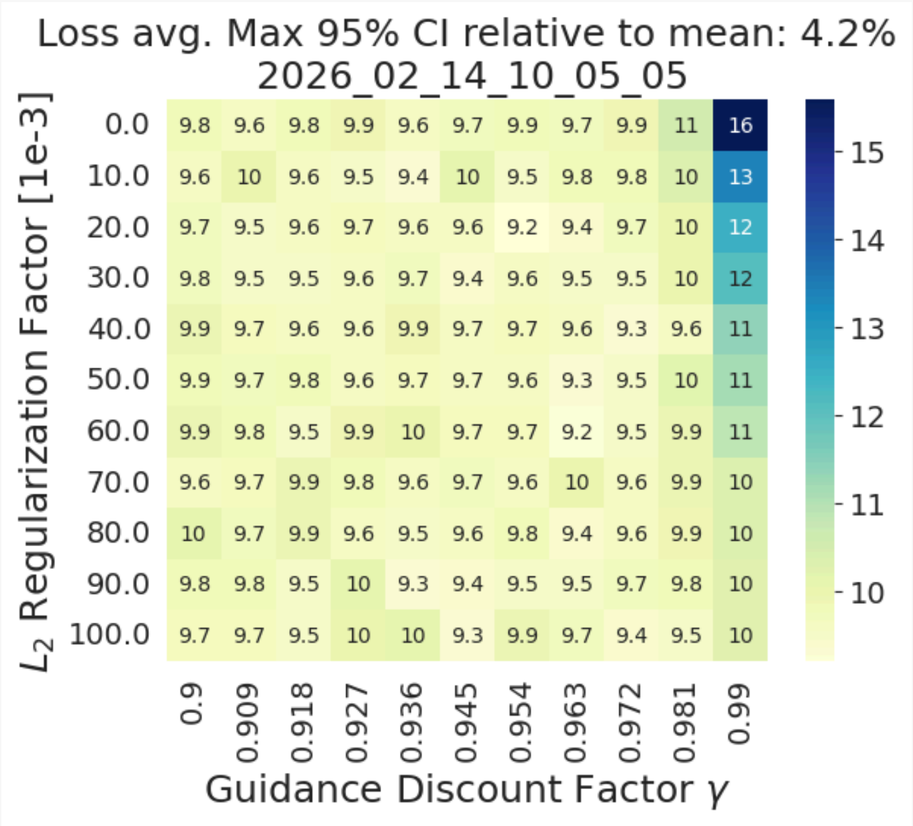
Few trajectories:

- High variance
- L_2 regularization and lower γ improve stability

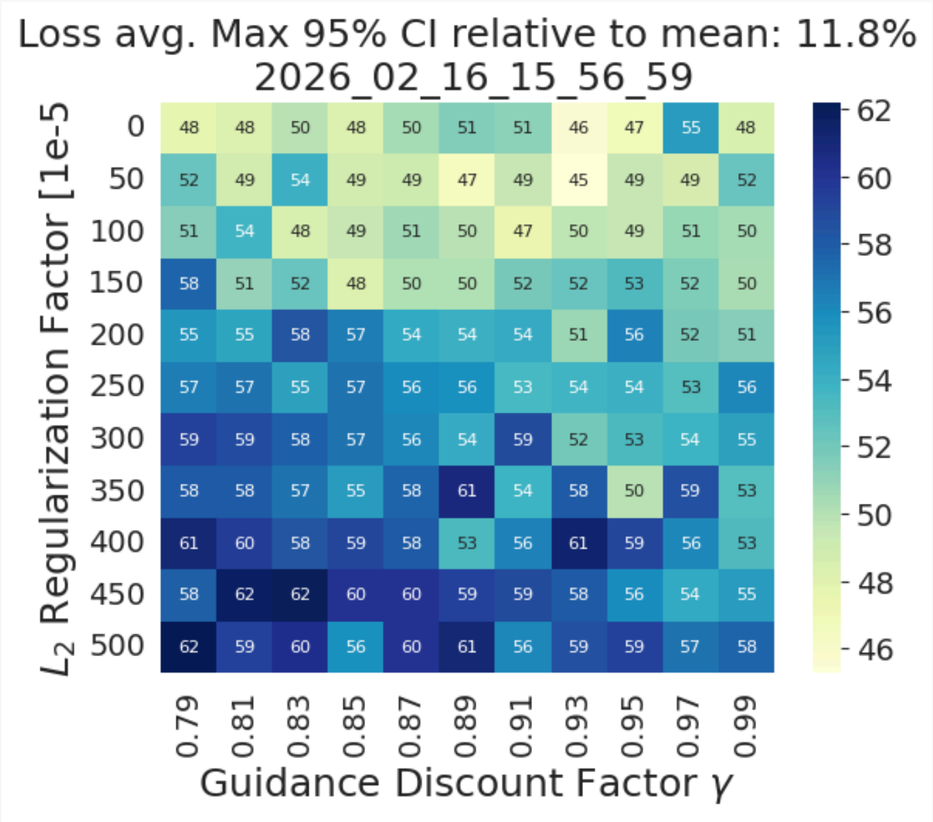
More data available:

- Regularization becomes unessential





Policy optimization with LSTD over 1000 repetitions



Policy evaluation with SARSA over 50 repetitions

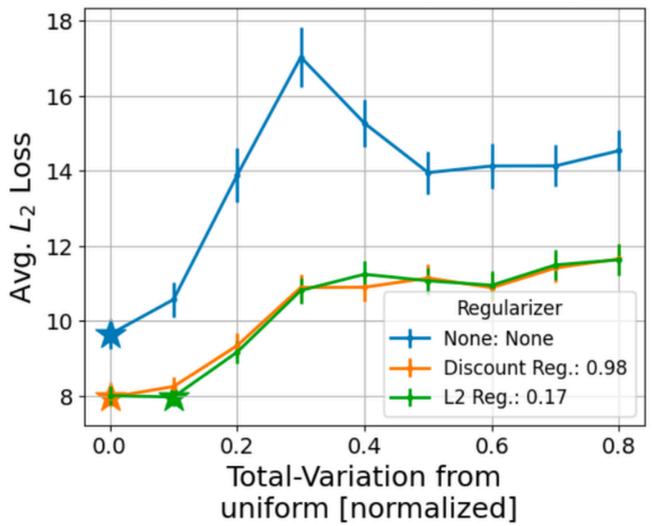
Performance depends jointly on the discount factor γ and the L_2 regularization strength.

- When γ approaches the evaluation discount γ_e, estimation becomes less stable and the loss increases.
- Moderate regularization and slightly reduced discount factors ($\gamma < \gamma_e$) lead to lower error.
- In policy optimization (SARSA), results are more variable due to limited repetitions and ϵ-greedy exploration.
- Higher γ values consistently produce higher loss.

We measure deviation from uniform distribution using **Total Variation distance.**

- When visitation is close to uniform: discount regularization is most effective
- When distribution becomes highly skewed: benefit decreases

→ States visited less often are less regularized.

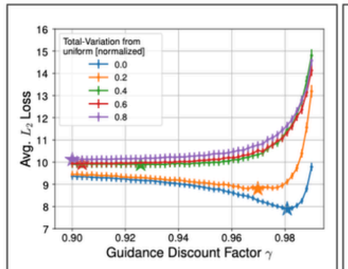


*On x-axis total variation from uniform (normalized)

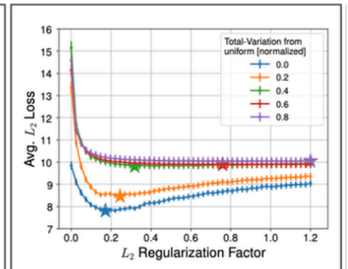
Slow mixing → higher correlation → higher variance

In slow mixing regimes:

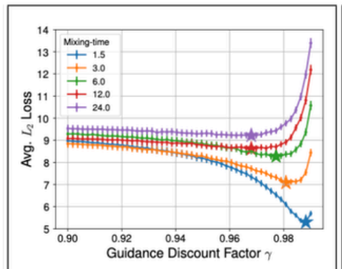
- Stronger regularization improves performance
- Both discount and L_2 help



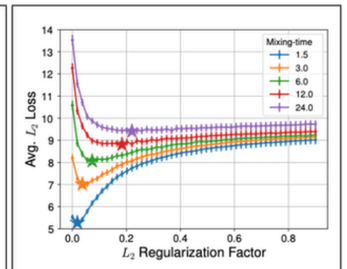
(a) L_2 loss vs. discount factor γ, for several total-variation distances from a uniform data distribution.



(b) L_2 loss vs. L_2 regularization factor, for several total-variation distances from a uniform data distribution.



(c) L_2 loss vs. discount factor γ, for several mixing-times τ.



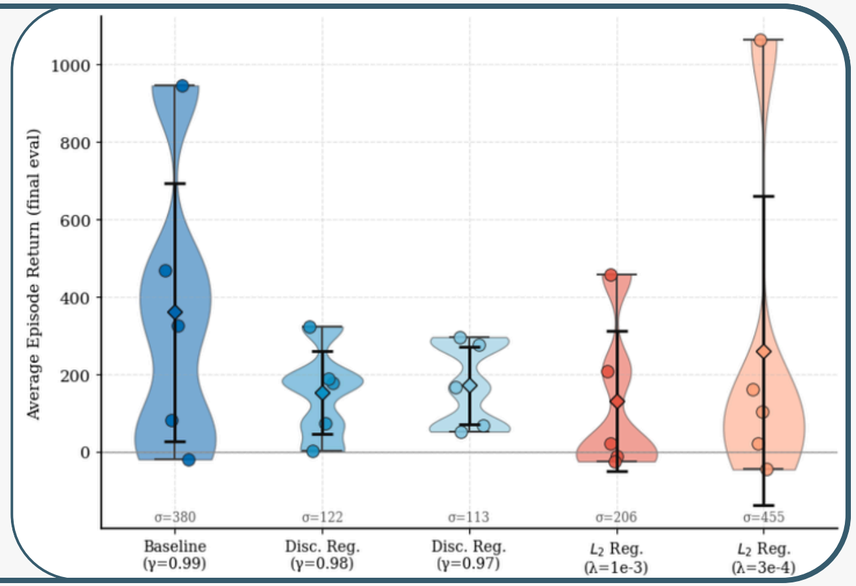
(d) L_2 loss vs. L_2 regularization factor, for several mixing-times τ.

DEEP RL EXPERIMENTS

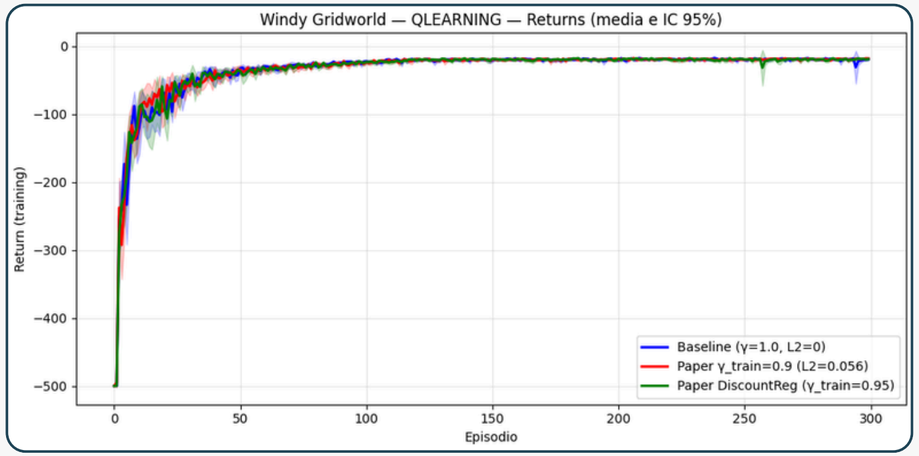
SETUP

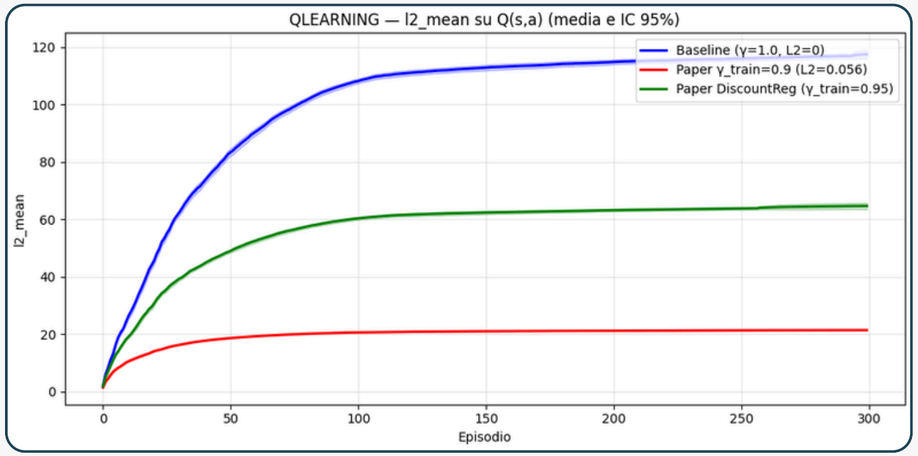
Continuous control tasks (MuJoCo)
Reduced training timesteps (25,000 and 5 seeds)
Algorithm: variant of Expected SARSA

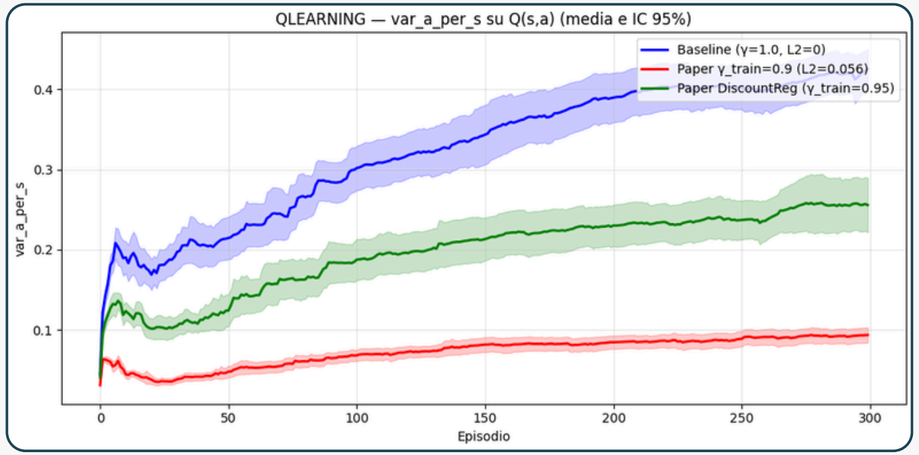
- Limited data → high variance across all configurations
- Baseline: shows the highest mean (driven by a single outlier seed)
- Lower discount factors reduce variance, suggesting a stabilizing effect of discount regularization.
- L_2 regularization appears instable across seeds

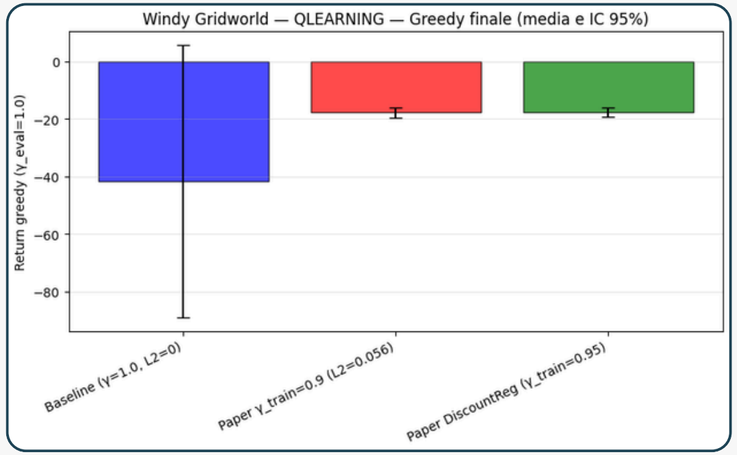


WINDY GRIDWORLD









CONCLUSIONS

EXPERIMENTS RESULTS

TABULAR:

- Lower γ improves stability in low-data regimes
- Both discount reduction and explicit L_2 decrease estimation error when trajectories are few
- Stronger regularization helps in slow-mixing environments (small spectral gap), reducing variance growth

DEEP RL:

- High variance due to limited seeds and training steps
- Lower γ shows reduced standard deviation across seeds (stabilizing effect)
- Explicit L_2 is more sensitive to initialization and less consistent
- Results are indicative but not statistically conclusive

LIMITATIONS

- Limited repetitions in tabular experiments
- Few seeds and reduced timesteps in deep RL

→ Statistical reliability is therefore limited

MAIN INSIGHT

Empirical evidence supports discount reduction as an effective regularizer, particularly in low-data and poorly conditioned settings

THANK YOU !